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- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Team ID-**
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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Legal Metrology officers manually inspect small declarations spread across every package surface; a missing photograph can be mistaken for a missing declaration, and an un-updated base PDF can be out of date.
- Guided multi-panel capture (camera or gallery, offline-resilient) feeds OCR and layout analysis; a deterministic rule engine issues a legally traceable finding citing the exact gazette clause.
- Six verdict states (PASS, FAIL, INDETERMINATE, NOT APPLICABLE, REVIEW, SYSTEM ERROR): “could not read” never becomes “missing” until the officer asserts full surface coverage.
- Applicability gates run before declaration checks, so exempt packages are never accused.
- The law itself is compiled offline from 48 gazette instruments into a human-reviewed, versioned, SHA-256 rulepack.
- No LLM makes compliance decisions — ordinary, auditable Python rules; officer corrections are append-only and attributed.

TECHNICAL APPROACH



- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Lane 1 (offline): gazette PDFs → law compiler → versioned, dated, SHA-256 rulepack — 48 instruments across 3 rule families (2011–2026).
- Lane 2 (per scan): guided photos → quality gates → OCR → layout/extraction → rule engine → six-state verdicts.
- One FastAPI service (localhost-capable) with PostgreSQL or a built-in local store, immutable local object storage, and optional S3.
- Clients: installable capture PWA, Android Field app (offline outbox with idempotent resumable sync), and a Next.js officer review workbench.
- Millimetre checks use an ISO ID-1 card as scale reference: corner marks → image-to-millimetre homography → panel dimensions in cm; mis-marked quads are refused and the engine abstains rather than guessing.
- E-commerce listings are checked under Rule 6(10) from shopper-visible listing text plus up to six screenshots.

FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea
 - Runs today on a laptop: portable Windows EXE for demos, or PostgreSQL plus local object storage for durable use — no S3 dependency.
 - Speed: 1.6 seconds per panel on ordinary CPU at the tested 1800 px cap.
 - Risk: extraction recall (~50%). Mitigation: a dedicated server workstream with a $\geq 90\%$ target — meanwhile the engine abstains (INDETERMINATE) instead of guessing.
 - Risk: real-device milestone gates (install, LAN pairing, airplane-mode capture, low storage) are still open. Mitigation: CI-built portable EXE and APK already published for device drills.
 - Production legal use requires qualified officer sign-off; the workbench supports corrections, re-evaluation, and reasoned overrides.
 - Deployment path: run now → controlled pilot → scale.

IMPACT AND BENEFITS



- **Potential impact on the target audience**
 - Beneficiaries: Legal Metrology officers, consumers, compliant brands, and MSMEs.
 - Validated on 140 real products (food, cosmetics, household, pet food) with 403 photographs and 2,700 rule evaluations.
 - 0 false accusations and 0 known violations silently passed in the recorded campaign; 20 defects found and fixed through adversarial testing.
 - Every finding retains the rulepack version, authority citation, evidence hash, and evaluation history.
 - Immutable local object storage with exportable signed PDF and DOCX reports.
 - The system answers confidently to roughly 10% character error, then stops answering rather than guessing.

- **Details / Links of the reference and research work**
 - Legal Metrology (Packaged Commodities) Rules, 2011 — compiled from 48 gazette instruments across 3 rule families (2011–2026).
 - Rule 6(10): online display of packaged commodities — the e-commerce listing checks.
 - Rule 7(2): MRP font-height minimum — enforced with perspective-corrected measurement from an ISO ID-1 scale reference.
 - Repository and evidence: github.com/akshajtiwari/lmpc-compliance-engine (README.md, docs/, docs/evidence/).
 - Preview release: v0.3.0-preview — portable Windows server and Android Field app, both published with SHA-256 checksums.